

Login Screen

Login Screen

<https://in.pinterest.com/pin/979181143996188940/>

Email Validation

1. Email field should not be empty.
2. Email format should be valid.
3. Leading and trailing spaces should be trimmed.
4. Maximum email length should be restricted. (Maximum 25 Characters)
5. Only valid characters should be allowed.

Password Validation

6. Minimum password length should be enforced. (Minimum 8 Characters)
7. Maximum password length should be restricted. (Maximum 15 Characters)
8. Must contain at least one number and one uppercase, lowercase letter and one special character and at least 8 or more characters
9. Password should be masked.
10. Show/Hide password option should work correctly.

Login Button Validation

11. Login button should remain disabled until mandatory fields are entered.
12. Button should not allow multiple clicks.
13. Loading indicator should appear during authentication.

Authentication Validation

14. Valid credentials should allow login.
15. Invalid credentials should display an appropriate error.
16. Locked users should not be able to log in.
17. Inactive users should not be able to log in.

Session Validation

18. Session token should be generated after successful login.
19. Session timeout should log out inactive users.
20. Logout should invalidate the session.
21. Users should not access pages after logout using the browser back button.

Forgot Password Validation

22. Forgot Password link should be visible.
23. Registered email should receive reset instructions.
24. Unregistered email should display an error.

Security Validation

- 25. Password should never be displayed in plain text.
- 26. SQL Injection attacks should be prevented.
Example : OR 1=1 - - should not bypass authentication.
- 27. HTTPS should be enforced.
- 28. Brute-force login attempts should be limited.
Failed Attempts: 5 -> Account Locked for 15 Minutes
(Account lockout after multiple failed attempts.)

UI Validation

- 29. Tab order should work correctly.
- 30. Error messages should be clear and user-friendly.
- 31. Required fields should be marked.
- 32. Login screen should be responsive.
- 33. Password visibility icon should function correctly.
- 34. Keyboard Enter key should trigger login.

Accessibility Validation

- 35. Labels should be associated with fields.
- 36. Screen readers should read field names correctly.
- 37. Color contrast should meet accessibility standards.
- 38. Keyboard-only navigation should work.

Audit Validation

- 39. Successful logins should be logged.
- 40. Failed login attempts should be logged.
- 41. User IP address should be captured.
- 42. Login timestamp should be recorded.

Critical Validation Checklist

- | | | |
|--|--|--|
| ☒ Email Required | ☒ Password Required | ☒ Valid Email Format |
| ☒ Password Masking | ☒ Invalid Credential Message | ☒ Forgot Password |
| ☒ Session Timeout | ☒ HTTPS Enabled | ☒ Account Lockout |
| ☒ Audit Logging | ☒ Responsive UI | ☒ Accessibility Compliance |